UNIT 2 · TOPIC 5(A)

Force Analysis on Tools and Implements

Syllabus topic: Force analysis on agricultural tools and implements: pull, draft, unit draft and power of farm equipment

The force system on an implement

An implement moving at constant speed is in equilibrium under three force groups:

1. Gravity (implement weight W).
2. Soil forces (useful soil resistance + parasitic forces).
3. The pull the tractor exerts through the hitch.

At steady state the vector sum of all forces = 0. For 3-D analysis, resolve the resultant soil reaction along three perpendicular axes: **L** = longitudinal component (opposes travel — the draft component), **V** = vertical component (up or down), **S** = side (lateral) component. Measuring the complete force on a mounted implement therefore needs an **L–V–S (three-axis) dynamometer**.

Useful soil forces	Parasitic forces
Resistance to cutting, pulverisation, inversion and displacement of the furrow slice — the work we actually want done.	Landside friction, sole (heel) friction, rolling resistance of gauge/transport wheels, bearing friction — they consume power without useful work.

$$\text{Total draft} = \text{useful draft} + \text{parasitic drag}$$

Good design minimises the parasitic share.

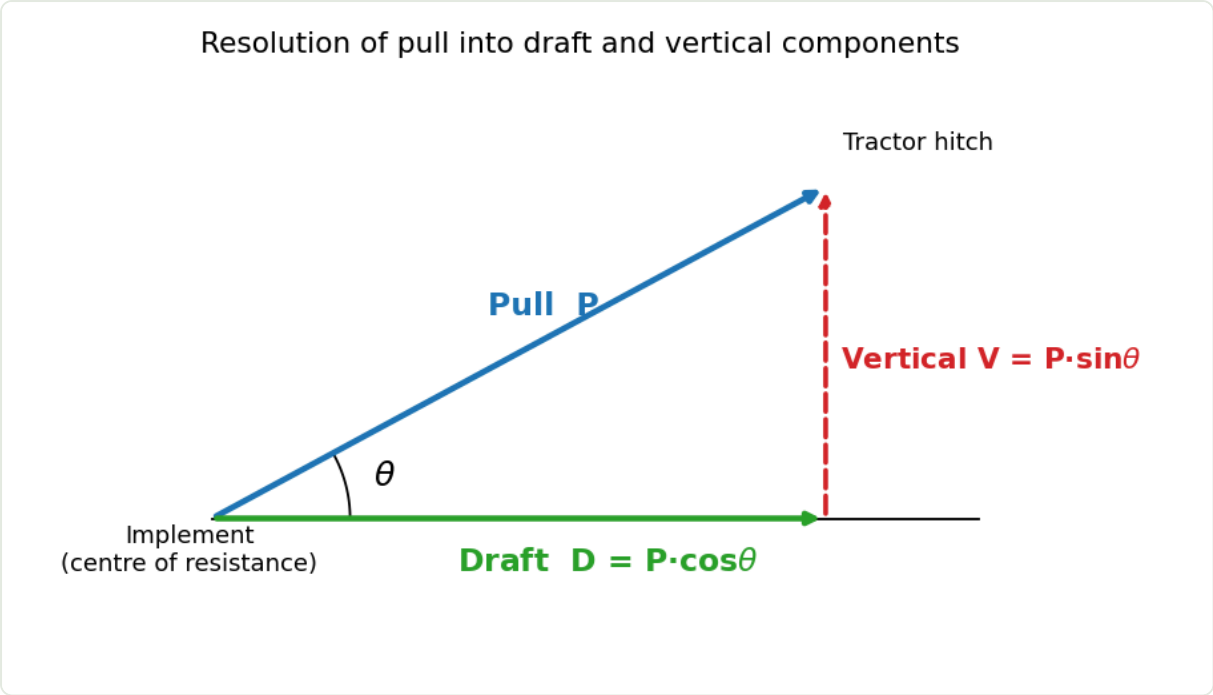
Pull, draft, side draft and vertical component

- **Pull (P):** total force on the implement along the line of hitch — generally inclined to the ground and possibly to the direction of travel.
- **Draft (D):** horizontal component of pull in the direction of travel. With vertical angle θ and lateral angle φ of the pull line:

$$D = P \cdot \cos\theta \cdot \cos\varphi; \text{ for pull inclined only vertically, } D = P \cdot \cos\theta$$

$$V = P \cdot \sin\theta$$

The tractor pulls the implement slightly **upward** along the hitch line. The equal and opposite reaction transfers weight onto the tractor's rear wheels (helping traction) while tending to lift the implement (reducing penetration).



Resolution of pull into draft and vertical components

- **Side draft:** the lateral component of pull, produced when the line of pull does not pass through the centre of resistance in plan view. It causes tractor "crabbing", heavy landside pressure, extra wear and steering effort — e.g., offset harrows, one-way plough, side-mounted mowers.

Centre of resistance and correct hitching

- **Centre of resistance:** the point where the resultant of all soil forces on the implement may be assumed to act. For a mouldboard plough bottom, take it at about **one-fourth of the width of cut from the landside** (shin side) and about **half the working depth** below the surface.
- **Correct hitching:** make the line of pull pass through both the hitch point (true or virtual) and the centre of resistance. Then no unbalanced moment acts on the implement → steady running, minimum draft, no side draft, uniform depth.

Vertical hitch (trailed implements)	Effect
Hitch too HIGH on the tractor	Steeper pull line → larger upward component on the implement → poorer penetration (plough runs shallow); less load on implement wheels.
Hitch too LOW	Flatter line → better penetration but more parasitic wheel load; an extremely low hitch can lift the tractor front (with mounted implements, risk of rearward overturn — see Topic 4b).

- **Horizontal (lateral) adjustment:** a lateral offset of the hitch from the centre of resistance creates side draft and a yawing moment. Correct it by shifting the hitch (cross-shaft/clevis) so the pull line passes through the centre of resistance in plan.
- **Mounted implements:** the lower links converge in plan and side view; their instantaneous intersection defines the **virtual hitch point**, which replaces the true hitch point in force analysis (basis of automatic draft control sensing — Unit 4).

Mouldboard vs disc tools — nature of soil reaction

Tool	Soil reaction and penetration
Mouldboard plough	Rigid bottom with vertical and horizontal **suction** (clearance behind the cutting edges); the soil force helps pull it in, so penetration is "self-acting" in normal conditions.
Disc plough/harrow	The freely rotating concave disc receives a soil reaction with an **upward** vertical component; it cannot "suck" itself in, so penetration must come from added mass (ballast) and disc/tilt-angle settings — hence disc ploughs are built heavy.

Measuring pull and draft

- **Trailed implements:** insert a dynamometer between tractor drawbar and implement — spring dynamometer (low accuracy, oscillation), hydraulic dynamometer (pressure gauge on a load cylinder) or strain-gauge load cell (electrical output, data logging — standard in research).
- **Mounted implements:** use a three-point-hitch dynamometer — either a frame ("quadrant"/EOR frame) inserted between tractor and implement, or instrumented (strain-gauged) lower links and top link; it resolves the L, V and S components.

Practical draft of a trailed implement:

1. Measure the pull with the implement working the soil (loaded).
2. Measure the towing resistance of the same implement running empty on the surface.
3. Subtract: practical draft = loaded pull - empty towing resistance. This separates soil-working draft from rolling resistance.

Practice MCQs

Q89. The centre of resistance of an MB plough bottom is commonly assumed at:

(a) ¼ width from landside, ½ depth
(b) ½ width, full depth
(c) At the share point
(d) At the heel of landside

Q90. For steady running with minimum draft, the line of pull must pass through:

(a) The rear axle
(b) Centre of resistance and hitch point
(c) The plough heel
(d) The mast top

Q91. Side draft occurs when:

(a) Speed is high
(b) The line of pull is laterally offset from the centre of resistance
(c) Depth is uniform
(d) Soil is sandy

Q92. Hitching a trailed plough too HIGH on the tractor drawbar generally causes:

(a) Deeper penetration
(b) Reduced penetration (plough runs shallow)
(c) No change
(d) More side draft only

Q93. A disc plough penetrates mainly because of:

(a) Suction of the bottom
(b) Added weight/ballast
(c) Sharp landside
(d) High speed

Q94. Which of these is a PARASITIC force on a plough?

(a) Soil cutting resistance
(b) Furrow slice acceleration
(c) Landside friction
(d) Soil inversion resistance

Q95. In L–V–S force analysis of a mounted implement, V denotes:

(a) Velocity
(b) Vertical force component
(c) Volume of soil
(d) Side force

Q96. Draft of a TRAILED implement is best measured with:

(a) Torque meter on PTO
(b) Dynamometer between drawbar and implement
(c) Fuel meter
(d) Tachometer

Q97. Draft of a MOUNTED implement is measured with:

(a) Spring balance on top link only
(b) Three-point hitch (strain-gauge) dynamometer
(c) Rope and pulley
(d) Prony brake

Answers

Q89 — (a) ¼ width from landside, ½ depth

Q90 — (b) Centre of resistance and hitch point

Q91 — (b) The line of pull is laterally offset from the centre of resistance

Q92 — (b) Reduced penetration (plough runs shallow) — *Higher hitch → steeper pull line → larger upward component $P \cdot \sin\theta$ on the implement → reduced penetration.*

Q93 — (b) Added weight/ballast

Q94 — (c) Landside friction

Q95 — (b) Vertical force component

Q96 — (b) Dynamometer between drawbar and implement

Q97 — (b) Three-point hitch (strain-gauge) dynamometer

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery (implement force analysis & hitching); ASAE Standard D497 (Agricultural Machinery Management Data — draft equation); Goryachkin's rational draft theory; Jagdishwar Sahay — Elements of Agricultural Engineering; NPTEL Farm Machinery & Equipment lectures.